Stock Pitch Challenge 2026

Hosted By

SemiAnalysis x Nanyang Capital

Submitted by Odyssey Capital

Nittobo

Tokyo Stock Exchange
(TYO: 3110)

Recommendation:

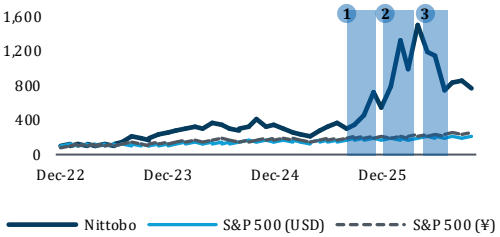
SELL



Figure 1: Key Company Data

RECOMMENDATION	SELL
Valuation Date	16 Oct, 2026
Target Price	¥2,392
Current Price	¥2,989
Upside / (Downside)	(20.0%)
Shares Outstanding	182.03 MM
Market Cap	¥563.7B
52 Week High / Low	¥6,580 / ¥1,412

Figure 2: 3Y Share Price Performance Against S&P 500 (Rebased to Dec 2022)

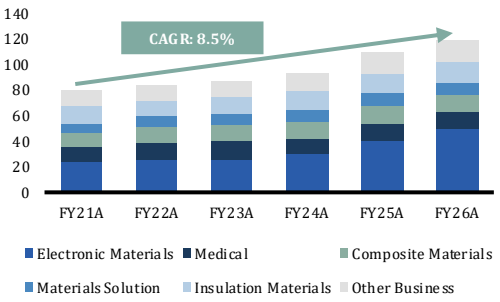


Source : Yahoo Finance

Figure 3: Key Events

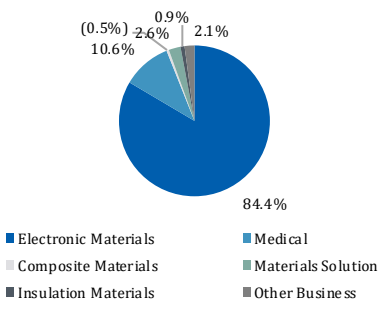
Period	Movement	Event
1: Oct - Dec 25	+97% to Nov, -28% to Dec	Bottoming out from margin compression / FX losses
2: Jan - May 26	+200% from Jan to May	Global supply shortage of T-glass and rotation into AI supply chain 'picks and shovels'
3: May - Jul 26	-59% from May to Jul	Conservative guidance, realising one-off gain in FY26A

Figure 4: Revenue By Segment (¥ B)



Source : Company filings

Figure 5: Operating Profit Mix (FY26A)



Source : Company filings

Figure 6: Special Glass Product Ladder

Product	Key property	Main use	Market Share
T-glass	Low-CTE	AI chip substrates	~90%
NER-glass	Low-Dk, 2nd gen	800G switches, servers	60–70%
NE-glass	Low-Dk, 1st gen	Server, switch boards	Leading
E-glass	Standard	Mainstream PCBs	Small

Source : Company filings, Team Analysis

Executive Summary

We initiate coverage on **Nitto Boseki Co., Ltd (TYO: 3110)** with a **SELL** recommendation based on a 6-month target price of **¥2,392**, implying **20.0% downside** to the last close of ¥2,989 as of 16th October 2026 (**Figure 1**). Our valuation is derived from a weighted mix of relative valuation (utilising the Sum-of-the-Parts methodology). Valuation was sense checked using a combination of Monte Carlo Simulation and sensitivity analysis in DCF method.

Nitto Boseki (Nittobo) is a Japanese manufacturer, with Special Glass for semiconductor being the key segment. Stock movement was volatile over the last one year compared to the S&P 500 index, as Nittobo's largest segment sits within the AI semiconductor packaging supply chain (**Figure 2**). The stock is driven by a few factors, including macro drivers like overall demand for Special Glass, USDJPY rate, and company specific drivers like margin and one-off gains (**Figure 3**). We believe that the recent downfall is not a full reflection of the industrial headwinds and Nittobo's inability to further improve margins. Overall, these drivers should lead to downside over the next 6 months. This represents **53.7% annualised IRR** from an entry price of ¥2,989.

Thesis 1: Electronic Materials Margin is Not Durable

The Street believes that EM margins stay in the mid-40s through FY29E, and expects the group OPM to rise due to the mix shifts toward EM. However, we believe that 1Q27's EM OPM is the peak as it benefits from the already-depreciated asset. Nittobo's massive capex plan and aggressive capacity targets result in additional costs that the Street is currently underestimating.

Thesis 2: The Order Air Pocket

The Street expects Nittobo's 2–3x capacity expansion to translate directly into earnings, with EM margins holding in the low 40s through FY29E. But we expect that capacity to land just as rationed customers stop over-ordering and use up the stock they built. That would repeat the FY23A and ABF slumps, with volumes falling on a cost base whose depreciation has more than doubled.

Thesis 3: Pricing Cannot Rescue Nittobo

The Street expects price increases tied to new capacity will offset the higher costs. However, we believe that T-glass pricing is now set on cost and investment, and management mentioned no more price increases driven by shortages, only cost pass-through.

Business Overview

Nittobo is a specialty-materials manufacturing company: Founded in 1898, Nittobo's roots began in textiles, but its current footing is in glass-fiber. In FY26A the group earned ¥118.2B of revenue with operating profit of ¥20.8B across the six segments (**Figure 4**). Revenue grew at 8.5% CAGR for FY21-26A, while its operating profit grew at 28.4% over the same time horizon, largely due to the pricing effect of Electronic Materials segment.

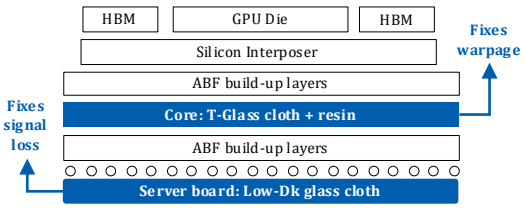
Glass-fiber is its bread & butter: Nittobo has exited most textiles, with ~66% of its sales now coming from glass-fiber. The Electronic Materials segment makes the glass yarn and woven glass cloth that reinforce copper-clad laminates (CCL) and integrated circuit (IC) package substrates. On 42% of group sales, it produced 84% of segment profit at a 39% margin in FY26A and took 86% of capex (**Figure 5**). Its high-end products are Special Glass: low-CTE T-glass, which keeps AI-chip package substrates from warping, and low-Dk NE-glass for high-speed server and switch boards. Nittobo supplies most of the world's T-glass, so group earnings now track one product's supply cycle.

Special Glass is the medium-term headline: Alongside a re-organisation from three divisions to six, which puts decisions closer to customers, management aims to harvest the investments it has already made and to build new revenue pillars for beyond 2030. The revised plan targets ¥155B of sales and ¥36B of operating profit by FY28E (up from ¥135B and ¥20B), backed by ¥120B of capex over four years (up from ¥80B).

The rest of the portfolio is mature: Management lists Medical as a focus business, but its numbers look like a mature one, earning a steady ~17% margin with little growth. Composite Materials loses money, and Insulation, Materials Solution and Other (mainly machinery) earn thin margins. Together these five segments made ¥3.6B of operating profit on ¥69B of sales. In practice, the ¥36B target rests almost entirely on Special Glass.

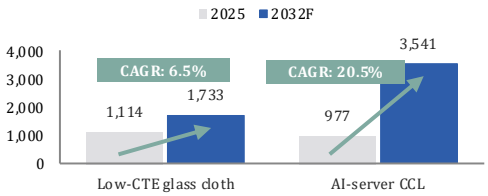
Nittobo negotiates prices individually by customer and product: T-glass prices were raised in FY27E and future adjustments focused on cost recovery rather than supply shortages. Sales are supported by rolling orders and customer forecasts, with no publicly disclosed take-or-pay or multi-year volume commitments. Nittobo primarily sells directly to laminate manufacturers. Its partnership with Nan Ya Plastics will outsource roughly 20% of glass-cloth weaving by 2027.

Figure 7: Glass Cloth in AI Server



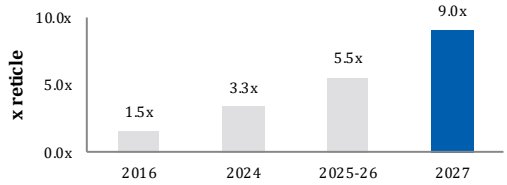
Source : Team Analysis

Figure 9: Market Size ($ M)



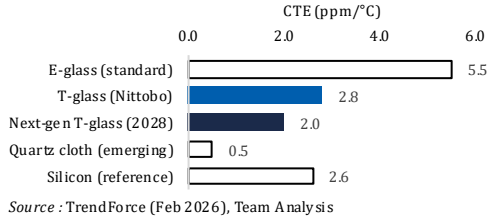
Source : LP Information

Figure 10: AI Package Interposers Size



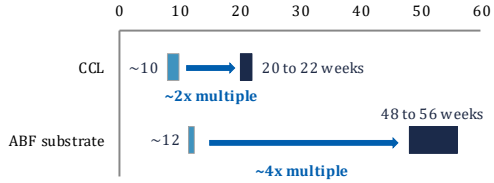
Source : TSMC, 3D InCites, TrendForce

Figure 11: Thermal Expansion



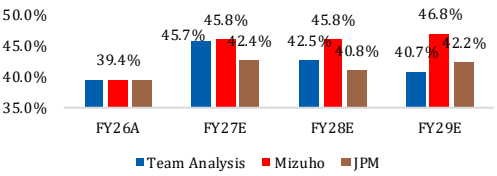
Source : TrendForce (Feb 2026), Team Analysis

Figure 12: Lead Time by Weeks



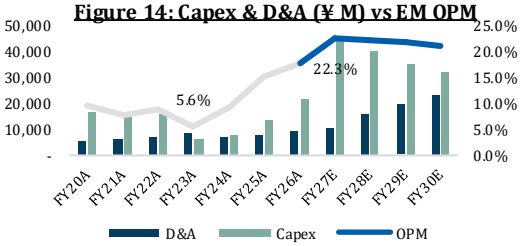
Source : TrendForce (Sep 2026)

Figure 13: Street vs Team Analysis OPM



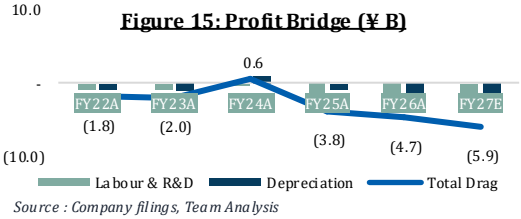
Source : Company filings, JP Morgan, Mizuho

Figure 14: Capex & D&A (¥ M) vs EM OPM



Source : Company filings, Team Analysis

Figure 15: Profit Bridge (¥ B)

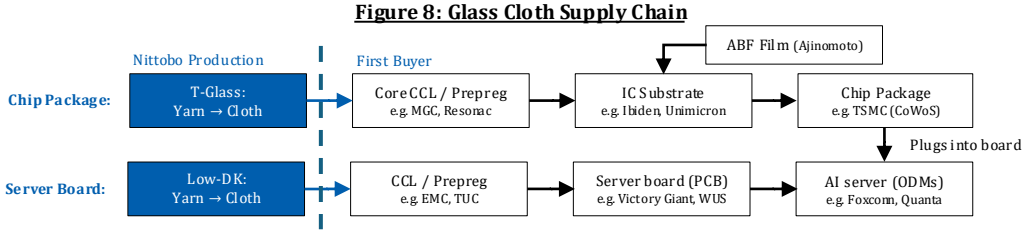


Source : Company filings, Team Analysis

Technology and Demand

Special Glass Ends Up in AI Hardware: Nittobo's Special Glass sits several steps from the chip. Yarn is woven into cloth and impregnated with resin into CCL or prepreg. T-glass laminate becomes the IC substrate core beneath a GPU or ASIC and its HBM, while low-Dk laminate becomes the server and switch boards those packages plug into. T-glass fixes warpage, so demand rises with AI chip units and, more sharply, with package size. Low-Dk fixes signal loss, so demand tracks server and switch shipments, compounded by each jump in speed **(Figure 7)**.

Figure 8: Glass Cloth Supply Chain



Market Sizing: According to LP Information, the global Low CTE Electronic Glass Fibre Cloth market size is predicted to grow from US$1.1B in 2025 to US$1.7B in 2032; representing a CAGR of 6.5%. At the same time, AI-server CCL demand is forecast to grow at 20.5% CAGR through 2032, creating a high-growth downstream opportunity for advanced electronic glass materials (**Figure 9**).

AI Packages are Getting Physically Larger: TSMC's CoWoS interposer has grown from 1.5x the lithography reticle in 2016 to 3.3x in 2024. It steps to 5.5x in 2026 and is planned at 9x for 2027, with substrates above 120 × 120 mm **(Figure 10)**. CoWoS capacity is expected to double from about 130k to 260k wafers a month between end-2026 and end-2028, a 41% CAGR. Each step adds substrate area and layers, so cloth demand is expected to grow faster than chip units.

Why AI Hardware Cannot Use Standard Glass: Silicon expands about 2.6 ppm per °C, far less than organic substrates, and T-glass closes that gap to within 0.2 ppm, keeping large packages flat during solder reflow (**Figure 11**). On boards, each doubling of SerDes speed, from 112G to 224G, raises signal loss, which low-Dk glass keeps within design limits.

Market Structure

A Capital-Intensive Industry: Weaving cloth is close to textile work, but drawing ultra-fine, uniform fibre from molten glass is not, and only a handful of producers can make M6-grade and higher yarn. Furnaces run nonstop at 1,600 to 1,700°C, each costs ¥0.5B to over ¥1.5B, and new high-end capacity takes over two years to build. With energy and depreciation largely fixed, profits typically swing with utilisation.

Tight Market with Early Substitutes: Glass cloth is now a bottleneck across the AI server supply chain. CCL lead times have stretched from 8 weeks to 20+, and ABF substrates from about 12 weeks to 48 to 56 **(Figure 12)**. The squeeze has spread to commodity grades: in China, the largest market, average e-cloth prices rose 148% year on year in 2Q26. Longer term, two technologies could replace glass cloth in parts of the stack. Quartz cloth targets the highest board grades, and glass core substrates swap the cloth core for a solid glass panel. However, both remain at early commercial stages.

Thesis 1: Electronic Materials Margin is Not Durable

Street View: Consensus treats the Electronic Materials (EM) margin as durable. Mizuho models EM OPM of 45-46% for FY27-29E and JPM around 40-42%. Consensus has group OPM expanding from 21.2% in FY27E to 27.3% in FY29E as EM grows as a share of the group (**Figure 13**).

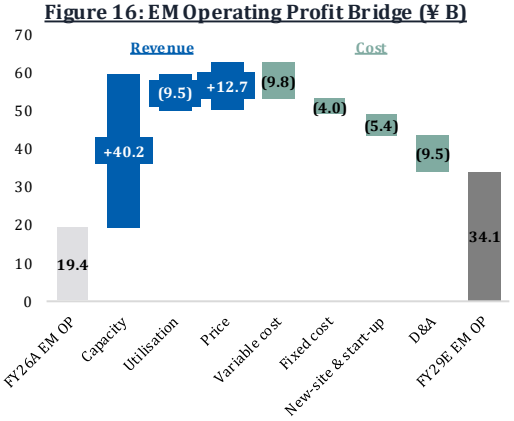
Variant View: We agree that EM revenue will continue to grow, but we disagree it will convert to profit at today's rate. EM OPM peaked at 48.2% in 1Q27, and the ~¥120B capex programme to roughly triple T-glass yarn capacity by end-FY28E brings a depreciation step-up that cost-linked pricing will not offset. We estimate **EM OPM of 40.7%** by FY29E versus Street's 42–46%.

Margin Follows the Capex Cycle, With a Lag: Nittobo has been here before. When Special Glass ran short in 2019, management spent beyond its mid-term plan and built a new furnace site in Taiwan, keeping capex at **~3x depreciation for three years**. In May 2021, management itself warned that profitability would decline as depreciation and personnel costs from the new sites came through. Depreciation rose 52% to a peak in FY23A, the same year group OPM troughed at 5.6%, a fall deepened by the energy-price shock. Margins recovered only after capex was cut and depreciation turned into a tailwind in FY24A. The current cycle follows the same pattern at a larger scale, as capex is now running at 4x depreciation on a ¥120B four-year plan (**Figure 14**).

Today's Margin Comes from Written-Down Assets: Around 70% of Nittobo's depreciable asset base has already been depreciated. Capex in FY23-25A ran at only ¥5.9-13.6B a year, roughly in line with depreciation. This changes in FY27E, as capex is guided at ¥45B against depreciation of ¥11.3B. Nittobo's **Medium-Term Management Plan** targets imply FY28E depreciation of about **¥15B** (EBITDA ¥51B and operating profit ¥36B), which is 61% above FY26A. Labour, R&D and depreciation have gone from a ~¥2B annual drag in the last expansion to ~¥6B in FY27E guidance (**Figure 15**).

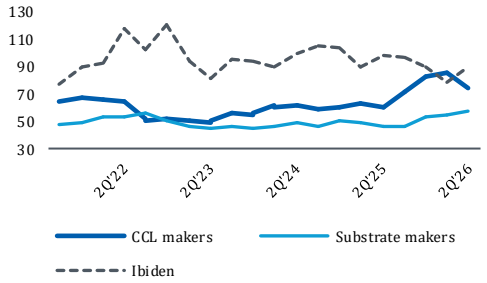
Investment Thesis

Figure 16: EM Operating Profit Bridge (¥ B)



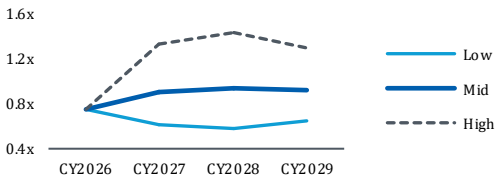
Source : Company filings, Team Analysis

Figure 17: Customer Inventory Days



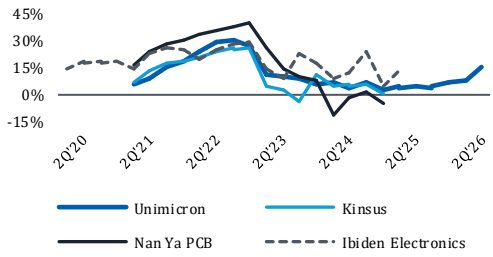
Source : Company filings, Team Analysis

Figure 18: T-Glass Supply/Demand (x)



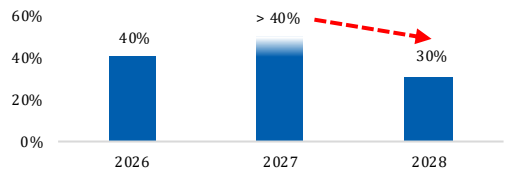
Source : Company filings, Team Analysis

Figure 19: ABF Operating Margins



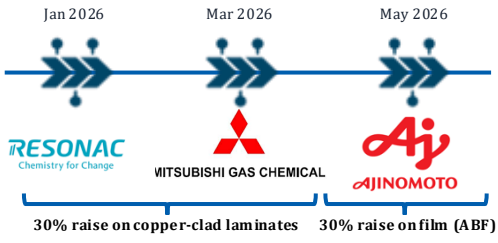
Source : Company filings, Team Analysis

Figure 20: Forecasts on T-Glass Supply Deficit from CY26-28



Source : Morgan Stanley

Figure 21: Price Increases From Downstream Peers



Source : Company filings, Team Analysis

Costs Arrive Before Revenue: Management's own profit bridge shows labour and R&D costs rising by ¥3.0B, ¥3.3B, and ¥3.9B in FY25-27E, while the depreciation drag grows from ¥0.8B to ¥2.0B. Guidance already reflects this. EM sales rise by ¥3.7B from 1H to 2H FY27E, yet EM operating profit falls by ¥0.2B, taking the segment margin (incl. intersegment sales) from 38.7% in 1Q to 34.1% in 2H (mgmt. cites depreciation, personnel and energy). Over FY26-29E, costs absorb ¥28.7B (66%) of the ¥43.4B in revenue-driven gains. Excluding the FY27E price increase, three years of capacity growth add only ~¥2B of profit (**Figure 16**).

New Lines Carry Start-up Costs: New furnaces must be heated for weeks before any output. High-silica T-glass runs at low yields until the process stabilises, and output cannot be sold until CCL makers, substrate makers, and chip designers qualify it. The greenfield Taiwan site and NE-to-T-glass line conversions, which take existing output offline, add further transition costs.

What It Would Take to Hold the Margin: Keeping EM OPM within the Street's 42–47% through FY29E requires one of two things. The first is near-full utilisation (against our 85% for T-glass and 92% for Low-Dk), which would add ~¥9B of revenue that drops almost entirely to profit. The second is a ~20% T-glass price increase in FY28E, against the ~5% pass-through we model. Management has ruled out the second, saying future increases will only recover costs or share the investment burden (May 2026), and the first requires the shortage to outlast a tripling of capacity. Theses 2 and 3 explain why neither is likely.

Thesis 2: The Order Air Pocket

Street View: Nittobo is a capacity story. The Street assumes that since every new furnace is already sold, any new capacity plans must realise into earnings. Capacity is guided to double by FY28E and reach roughly 3x FY26A levels by FY29E. The company targets ¥36B of group operating profit for FY28E, and brokers keep EM margins in the low 40s through FY29E. On that view, ~25x FY27E earnings must be a fair price for lasting growth.

Variant View: Customers on rationed supply are ordering ahead of use and building stock. When Nittobo's and its rivals' capacity lands together in FY28-29E, lead times shorten, customers stop ordering ahead and switch to using stock. Orders then fall below end demand for several quarters even if AI demand keeps growing. We call this an *order air pocket*. Nittobo sells at prices fixed once a year, with no volume commitments, so nothing protects it. The drop would also hit a bigger cost base: EM depreciation rises from ¥7.2B (FY26A) to ¥16.7B (FY29E).

It Has Happened Before: In FY23A, record orders driven by "anticipated demand" turned into "supply chain inventories". Nittobo's yarn and cloth sales fell 32% in three quarters, and quarterly operating profit swung from +¥1.8B to –¥1.5B. Excluding the 2022 energy shock and a deliberate production cut, about 50–55% of lost sales came straight off profit (**Figure 4**).

The Pattern Is Repeating: Nittobo's laminate customers held about 85 days of inventory in 1Q26. That is above the ~67-day peak that came before the FY23A fall. TUC reached 119 days. In 2Q26, their inventory grew 117% year on year, against revenue growth of 92%. Meanwhile, Nittobo's own inventory is up ¥13.2B since March 2025. Finished goods are back at their March 2023 level. With lead times stretching from about 8 to 20 weeks, customers must order roughly 20 weeks ahead and hold buffer stock. When new capacity brings lead times back to 8 weeks, about 12 weeks of orders disappear once, even if end demand keeps growing. (**Figure 17**).

Nittobo's Fukushima plant starts production in Jan–Mar 2027, and its Taiwan capacity doubles by March 2028. In the same window, Grace Fabric, Taiwan Glass, Taishan, Kingboard and Fulltech are adding capacity for low-CTE glass. Our illustrative supply-demand estimate suggests the shortfall narrows from about 25% in 2026 to under 10% by 2028, though the range is wide. It doesn't need to become a glut. Once the shortage eases, customers will stop ordering ahead (**Figure 18**).

The ABF precedent: ABF substrates sit one step above Nittobo in the supply chain, and they followed this exact path in 2021–24: (a) customers placed orders as far out as 2030 driven by fears of a shortage, (b) AT&S received €446M, and Intel reserved production lines for five years, and finally (c) customers cut orders while they worked through their stock. This caused OPM to fall by 25–50 points, because customer funding protected neither volumes nor margins. This matters because Nittobo plans to have customers share the cost of future expansion (**Figure 19**).

Thesis 3: Pricing Cannot Rescue Nittobo

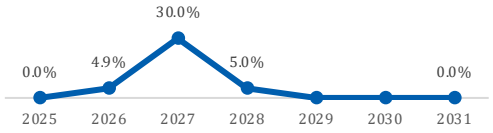
Street View: Following aggressive **price increases** across downstream materials suppliers (**Figure 21**), JPM report **forecast T-glass price increases of 27.7%** through FY27E. The Street therefore expects persistent supply scarcity and Nittobo's market dominance to sustain pricing growth.

Variant View: With a concentrated customer base and Vlex qualification imminent, Nittobo's room to raise prices without jeopardising relationships is constrained in the near term. In 1Q27, management explicitly ruled out further supply/demand-driven price revisions, prioritising capacity expansion and market share.

Priced Unlike A Monopoly: With Nittobo trading at a premium (**Figure 30**), the Street values the Electronic Materials segment as a monopoly capable of pricing leverage and industry-leading earnings growth. Yet, management has affirmed that price rises will only be capacity-linked rather than arising from scarcity premium. We foresee that these price increases will only recover part of their new depreciation on capex, but will not support the margins expected by the Street.

Thesis & Valuation

Figure 22: T-Glass Pricing Forecasts



Source : Team Analysis, JP Morgan, Mizuho

Figure 23: Revenue Build Methodology (EM)

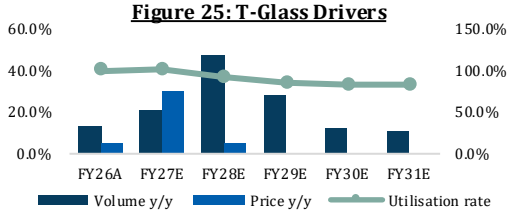


Applied the same methodology for both T-glass and Low-dk revenue

Figure 24: Cost Assumptions (EM)

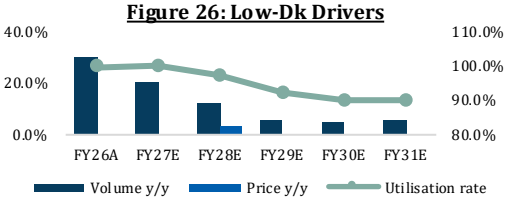


Figure 25: T-Glass Drivers



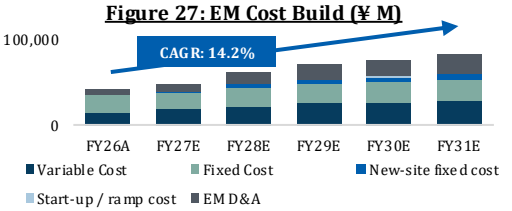
Source : Team Analysis, JP Morgan, Mizuho

Figure 26: Low-Dk Drivers



Source : Team Analysis, JP Morgan, Mizuho

Figure 27: EM Cost Build (¥ M)



Source : Company filings, Team Analysis

Figure 28: Football Field (¥)

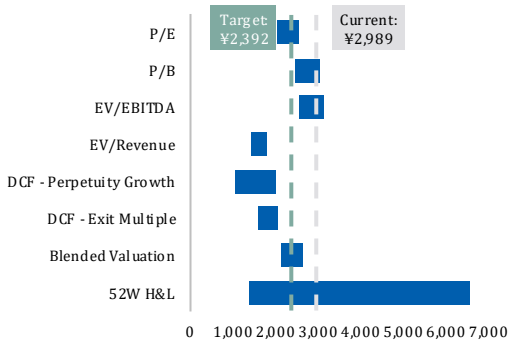
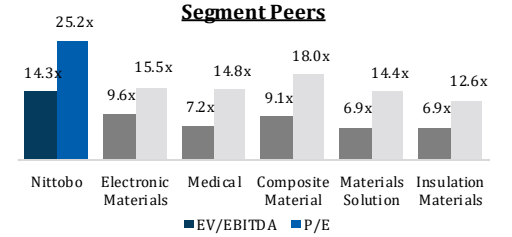
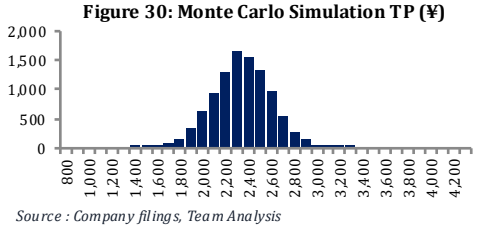


Figure 29: Premium Valuation to Segment Peers



Source : Bloomberg, Yahoo Finance

Figure 30: Monte Carlo Simulation TP (¥)



Source : Company filings, Team Analysis

Direct Material Substitution: Glass cloth capacity sells into a concentrated chain of laminators and substrate makers, including Resonac and Mitsubishi. These are the same customers qualifying Nittobo's **next-gen Vlex commercial rollout targeted for 2028**. Further price increases lower the economic barrier for customers to qualify direct, high-performance alternatives. Notably, quartz cloth from competitors like Shin-Etsu and Asahi Kasei outperforms T-glass on key metrics like dielectric constant (Dk) and dissipation factor (Df) (**Figure 11**), constrained only by its multiples-higher cost premium. By narrowing this price gap, Nittobo actively accelerates customer adoption of quartz cloth options over T-glass.

Architectural Shift Towards Glass Core Substrates: On the packaging layer, glass core substrates (GCS) currently cost more per unit area than traditional organic substrates, but panel-level processing is projected to reach parity with high-end organic substrates by 2028. With initial production from Intel and Samsung expected in late 2027, a price-induced nudge towards costlier GCS would accelerate the replacement of woven-glass core on which ABF layers are built, reducing overall demand for T-glass.

Revenue Build & Cost Assumptions

Nittobo Reports Revenues Across Six Segments: Electronic Materials (EM), Medical, Composite Materials, Materials Solution, Insulation Materials and Other Business. Since EM generates the majority of segment operating profit and is where our thesis lies, we built its revenue bottom-up. The remaining five segments are mature businesses and were forecast top-down: FY27E growth is anchored to company guidance and recent run-rates, and growth and margins are held near historical levels thereafter.

Revenue: We disaggregated EM into T-glass, Low-Dk (NE/NER-glass) and E-glass/others, with T-glass and Low-Dk revenue modelled as a function of capacity, utilisation and price (**Figure 23**). Capacity follows Nittobo's announced furnace expansions, under which management has indicated T-glass yarn capacity will **roughly triple** by end-FY28E. It was converted into volume using a utilisation rate that stays at full capacity in FY27E before easing as new supply outpaces incremental demand. Volume was then multiplied by a price index. For T-glass, this reflects the hike implemented at the start of FY27E, which we assume at around 30% in line with sell-side estimates as the company has not disclosed its size (**Figure 25**). A modest capacity-linked increase follows in FY28E, with flat pricing thereafter, consistent with management's stance that it will not raise prices on supply-demand conditions alone. Low-Dk pricing was held flat in FY27E, as management has ruled out revisions during the year, while E-glass/others decline gradually as Nittobo shifts resources towards special glass (**Figure 26**)

Costs: We built EM operating profit from its cost structure so that margins reflect the fixed-cost burden of the new capacity (**Figure 27**). Variable costs were modelled as a share of revenue at constant FY26A prices, so price increases flow directly to profit while volume growth carries its associated cost. Fixed cash costs grow with the rising personnel and energy costs flagged by management, and **new-site** and **start-up costs** are layered on as each line comes online. Depreciation was taken from our PP&E schedule, where capex in line with the ¥120B Medium-Term Plan budget is depreciated as it enters service. As a result, EM margins peak in FY27E and compress thereafter as depreciation and new-site costs grow faster than revenue.

Valuation

Our team has a **SELL** rating with a 12-month price target of ¥2,392, representing a 20% downside to the last close of ¥2,989 on 16 October 2026. Our price target is derived from an equally weighted blend of P/E, EV/EBITDA, P/B, and EV/Rev, with each segment valued against its own peer group on a sum-of-the-parts basis (**Figure 28**). As EM sits within the cyclical semiconductor value chain and the next 12-18 months of earnings are central to our thesis, we favoured forward multiples over intrinsic valuation, which depends heavily on long-term assumptions beyond the current capacity cycle.

P/E and EV/EBITDA are the sell side's preferred multiples for Nittobo and together bracket the investment cycle: EV/EBITDA strips out rising depreciation from new capacity, while P/E captures its full impact on earnings. EV/Rev values EM on revenue rather than peak-cycle margins, and P/B reflects Nittobo's asset-heavy base, where book value grows with capex. We cross-checked intrinsic value with a perpetuity-growth and exit-multiple DCF (**Appendix 3A**). A 10,000-iteration Monte Carlo simulation shows 99% of outcomes below the current price (**Figure 30**). Nittobo trades at a premium to every segment peer group on both P/E and EV/EBITDA (**Figure 29**), reflecting confidence in its near-monopoly in T-glass and 60–70% share in Low-Dk glass. Rather than assume this premium persists, we value each segment at its peer median (**Figure 32**). The exception is EM, where a **20% premium** to the glass fibre median on P/E, EV/EBITDA and EV/Rev (**Figure 31**) recognises Nittobo's T-glass lead and position as primary supplier.

Figure 31: Electronic Materials Comps

CY27	PE	EBITDA	Rev
Minimum	12.2x	6.7x	0.9x
1st Quartile	13.2x	7.8x	1.2x
Median	15.5x	9.6x	2.3x
Mean	20.1x	14.3x	5.4x
3rd Quartile	16.9x	10.3x	4.6x
Maximum	47.4x	37.3x	21.0x

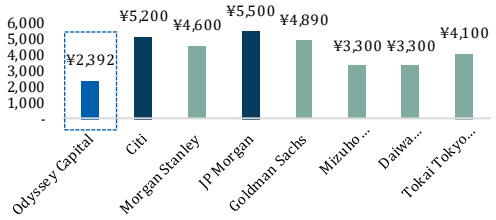
Source : Bloomberg, Company filings, Team Analysis

Figure 32: Multiples by Segments

	PE	EBITDA	Rev
Medical	14.8x	7.2x	0.9x
Composite Materials	18.0x	9.1x	2.9x
Materials Solution	14.4x	6.9x	0.9x
Insulation Materials	12.6x	6.9x	1.4x
Other Business	12.3x	7.2x	0.7x
(+) Corporate adjustments (unallocated OP)		9.2x	

Valuation, Risks & Catalysts

Figure 33: Consensus Target Price (¥)



Source : Company filings, Team Analysis

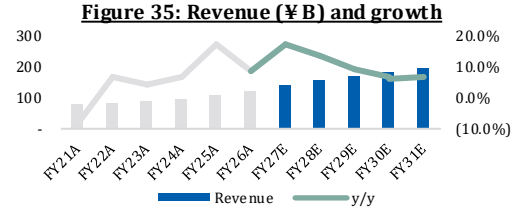
Figure 34: Scenario & EM Premium Sensitivity

EM Premium to Peer Median Multiples

TP (¥)	0%	10%	20%	30%	40%	50%
Bull	¥2,320	¥2,460	¥2,600	¥2,740	¥2,879	**¥3,019**
Base	¥2,145	¥2,269	**¥2,392**	¥2,515	¥2,638	¥2,761
Bear	¥1,952	¥2,057	¥2,162	¥2,267	¥2,372	¥2,477

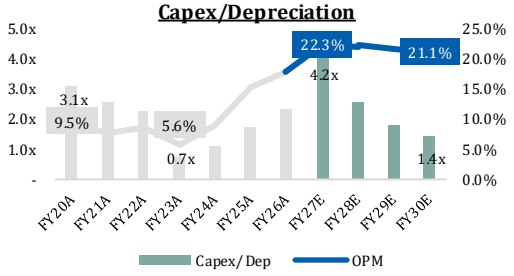
Source : Team Analysis

Figure 35: Revenue (¥ B) and growth



Source : Team Analysis

Figure 36: Profitability vs Capex/Depreciation



Source : Company filings, Team Analysis

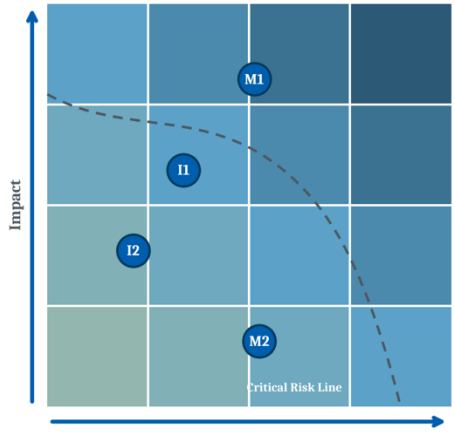
Figure 37: TP Sensitivity to FY28E T-glass Utilisation (UTR) and Price

FY28E T-glass Price Index y/y

FY28E T-glass UTR	1.0%	3.0%	5.0%	7.0%	9.0%
84.0%	¥2,258	¥2,280	¥2,302	¥2,324	¥2,346
88.0%	¥2,301	¥2,324	¥2,347	¥2,370	¥2,393
92.0%	¥2,344	¥2,368	**¥2,392**	¥2,416	¥2,440
96.0%	¥2,386	¥2,411	¥2,436	¥2,461	¥2,487
100.0%	¥2,429	¥2,455	¥2,481	¥2,507	¥2,533

Source : Team Analysis

Figure 38: Risk Probability and Impact Matrix



Source : Team Analysis

This premium is set well below the multiples the market and the sell side currently assign, as we expect Nittobo's advantage to narrow while Taiwan Glass, Nan Ya, Unitika and Chinese producers gain spec-in and new supply rebalances the market. We applied the same premium across P/E, EV/EBITDA and EV/Revenue, reflecting our view that once the shortage eases, Nittobo's earnings will be priced closer to other glass fibre producers.

Implying a Short: The Street remains bullish on Nittobo, with every covering broker setting a target price above the current share price **(Figure 33)**. Taking the lowest, ¥3,300 from Mizuho and Daiwa, as the risk to our call, its 10.4% upside compares with our 20.0% downside, **a reward-to-risk of about 2:1**. The gap stems largely from how EM is valued. JP Morgan applies an EV/EBITDA multiple of 20x to EM in its sum-of-the-parts, more than double the 9.6x peer median, while Mizuho applies a P/E of around 23x to the group. In our base case, the current price of ¥2,989 is only justified with a ~69% premium to the peer median, about 26x P/E and 16x EV/EBITDA on EM, and even our bull case needs ~48%, where it's bordered by red **(Figure 34)**. The Street's multiples assume today's shortage margins will last, while we expect new capacity to bring depreciation that dilutes EM returns from FY28E, supporting our SELL rating.

Financial Analysis

Top-line: Using the bottom-up revenue build methodology, Nittobo's revenue from FY26A to FY31E grows at a 10.5% CAGR compared to historical CAGR of 8.5% from FY21A to FY26A, while EM revenue grows at 14.0% CAGR from FY26A to FY31E vs 16.1% CAGR from FY21A to FY26A (**Figure 35**). New commercial furnace lines in Fukushima ramp output through late 2027, forcing a strategic shift from price increases that pass on expansion costs to volume defence, as the shortage eases and competition increases.

Cost & Profitability: Cost of Goods Sold (COGS) drops from 59.8% of revenue to 57.5% of revenue from FY26A to FY31E, but rises at a CAGR of 9.6% in FY26-31E vs 5.9% in FY21-26A. This is largely due to the costs associated with the new commercial furnace lines in Fukushima, as expansion requires capital expenditure in building fabrication facility, start-up cost in heating up new furnaces, and depreciation costs from massive capex. Historically, profitability has been low in years where depreciation was heavy relative to capex as Capex/Dep ratio was 0.8x, and the operating margin was as low as 5.6% (**Figure 36**). We expect the same trend to continue as Nittobo announced a massive four year ¥120B capex plan, lowering the capex/depreciation ratio to 1.4x and OPM to 21% from 22.3% (**Figure 36**).

Risks & Catalysts

[I1]: Quartz Substitution

Our base case eases Low-Dk utilisation to 92% by FY29E as quartz takes the top board grades. At ~40x the cost of E-glass, however, quartz could stay confined to a few boards for longer Low-Dk is Nittobo's biggest product, at about half of EM revenue, but the risk falls mainly in FY29E, beyond the year that drives our target price.

Mitigant: Quartz is already used in parts of Rubin and moves to M9+ boards for Rubin Ultra in 2027, while Nittobo's next-generation glass, Vlex, only launches in 2028. Asahi Kasei's quartz qualification by March 2027 will show which way it goes.

[I2]: Competitor Entry

If rivals cannot make low-CTE cloth at yield, Nittobo's ~90% T-glass share could hold longer, and scarcity pricing could return above the ~5% increase we model for FY28E.

Valuation Impact: Each 2pp of extra FY28E T-glass price adds about ¥24 (1.0%) to our target price; even a 9% increase lifts it only to ¥2,440 **(Figure 37)**.

Mitigant: Taiwan Glass has been in T-glass mass production since 1Q26, and Prismark expects Nittobo's share of 2028 T-glass capacity to fall to about 55% (Figure 2B). Any CCL maker naming a second source is the signal to watch.

[M1]: AI Demand Cycle

Hyperscaler spending that keeps beating forecasts would fill the new lines quickly and keep T-glass utilisation near full instead of easing to our 85% for FY29E, the main upside risk to our view (Figure 3A).

Valuation Impact: Full FY28E T-glass utilisation, against the 92% we model, would raise our target price by 3.7% to ¥2,481, still 17% below the current price **(Figure 37)**.

Mitigant: Cloth orders follow the growth in capex, not its level, and top-nine CSP capex growth slows from 106% in 2026 to 60% in 2027 and 12% in 2028 (Figure 2D). Supply is also outrunning demand, with T-glass capacity up ~2.2x over 2026 to 2028E against ~2.0x for CoWoS. FY28E guidance in May 2027, after Fukushima starts up, will be the test.

[M2]: Yen Exposure

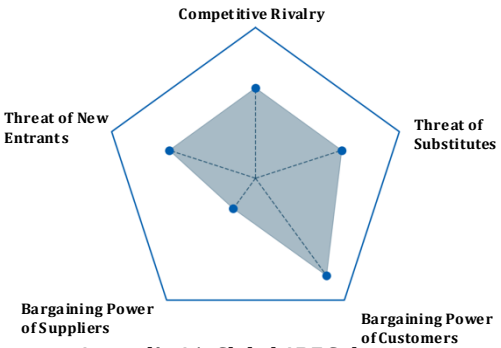
A weaker yen, which helped the August guidance upgrade, would lift reported profit against our call. Though there isn't company disclosure, we estimate ¥0.13B to ¥0.23B of OP per ¥1 move against the dollar, or 0.4% to 0.8% of FY27E OP guidance.

Mitigant: The exposure is small and cuts both ways, as further yen strength after July's US-Japan intervention would remove a recent tailwind.

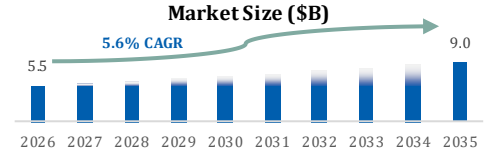
Appendix 1: Company

Key Financials	FY22A	FY23A	FY24A	FY25A	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue	84,049	87,528	93,250	109,033	118,226	138,627	157,006	170,966	181,887	193,564
Revenue Growth (% YoY)	6.8%	4.1%	6.5%	16.9%	8.4%	17.3%	13.3%	8.9%	6.4%	6.4%
Operating Profit	7,268	4,880	8,387	16,444	20,818	30,942	34,719	37,060	38,326	39,964
EBITDA	14,350	13,088	15,553	24,386	30,120	41,700	50,141	56,708	61,198	65,779
Net Income	6,520	2,773	7,286	12,837	41,770	21,555	23,974	25,588	26,474	27,608

Appendix 1A: Porter's Five Forces

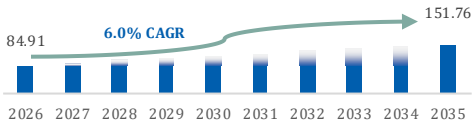


Appendix A1: Global ABF Substrate Market Size ($B)



Source : Market Growth Reports, Team Analysis

Appendix A2: Printed Circuit Board Market Size ($B)



Competitive Rivalry (Moderate): Nittobo still leads both of its key glass cloths, with ~90% of T-glass and ~70% of NE-glass, but the two markets are moving in different directions. Low-Dk already has four other suppliers, and Taiwan Glass is expanding its high-grade lines from 4 to 12. In T-glass, Taiwan Glass has been in mass production since 1Q26. Prismark expects Nittobo's share of 2028 capacity to fall to ~55% as rivals grow by nearly 2.5x.

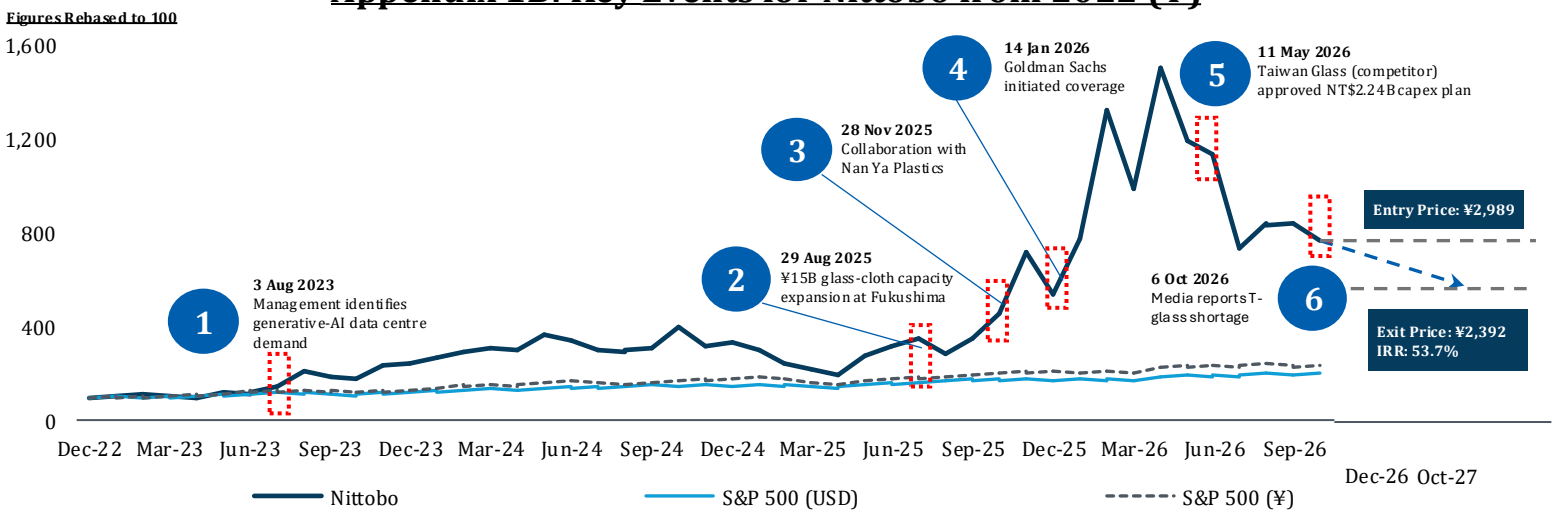
Threat of New Entrants (Moderate): Structural barriers to entry remain high. Developing high-end specialty glass drawing capacity requires massive upfront capital expenditure, proprietary furnace designs, and a rigorous 12-to-18-month qualification process with major CCL fabricators and tech OEMs. But these barriers are semi-permeable, not impenetrable. Taiwanese rivals moved from sampling in Aug-24 to spec-in by Nov-25, and the shortage has encouraged customers to approve them faster.

Bargaining Power of Suppliers (Low): Nittobo faces minimal supplier pressure regarding raw material inputs. Primary raw ingredients, such as high-purity silicas and standard mineral compounds, are widely available commoditised inputs sourced from diverse global suppliers, enabling Nittobo to secure favourable procurement terms and mitigate severe input cost shocks.

Bargaining Power of Customers (Moderate-High): Buyers possess notable leverage as Nittobo's downstream customer base is concentrated among a select group of mega-cap substrate manufacturers and CCL fabricators. These sophisticated buyers command significant volume purchasing power, but their ability to extract aggressive price concessions is constrained because end-user tech OEMs explicitly specify Nittobo's proprietary glass cloth grades in their blueprints.

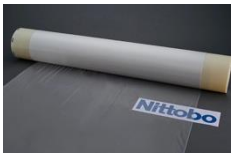
Threat of Substitutes (Moderate): The structural threat of substitution is substantial over a multi-year horizon. Quartz cloth already competes with Low-Dk on the highest-grade boards, including parts of NVIDIA's Rubin, though at ~40x the cost of E-glass it should stay a niche. Glass-core substrates would replace the T-glass core, but production is not expected before late 2027–2030.

Appendix 1B: Key Events for Nittobo from 2022 (¥)

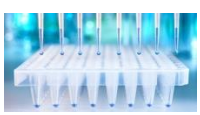


Appendix 1C: Product Offerings

Electronic Materials
EM segment sells T-glass and Low-Dk glass, which are ultra-thin glass fibre, spun into yarn and woven into cloth. The cloth is soaked in resin and used as the rigid, heat-stable core of circuit boards and chip-package substrates



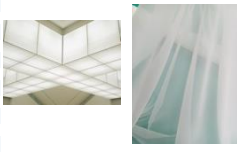
Medical
Medical sells raw materials and reagents for in-vitro diagnostics, essentially blood test. It sells to diagnostic makers and hospital labs, not directly to patients.



Composite
Glass fibre sold as a reinforcement for plastics (FRP). Think of glass strands or chopped glass mixed into resin so a part gets stronger and stiffer. It is used in things like car components, equipment housings and industrial parts.



Materials Solution
MatSol sells industrial and constructional glass-fibre textiles (heat-resistant and building-material cloths, tapes, etc) and special chemicals, such as functional polymers.



Insulation
Glass wool, the fluffy pink or yellow insulation in house walls and ceilings, plus industrial and equipment insulation, sold to Paramount Glass Manufacturing.



Other
Mainly the trading company Nittobo Global Trading, formerly Soyo. It resells glass cloth, so its sales track the EM intersegment flow (about 20% of EM gross sales)



Appendix 2: Industry

Figure 2A: Profit Pools and Roles by Layer

Layer	Key players	Share	Role	OPM
Yarn / cloth	Nittobo, Taiwan Glass	~90% T-glass	Sells	**39%**
CCL / core	EMC, MGC, Resonac	~40% (EMC)	Buys	27%
ABF film	Ajinomoto	~95%	Sells	54%
IC substrate	Ibiden, Unimicron	Moderate	Qualifies	28%
CoWoS	TSMC, ASE, Amkor	~90% (TSMC)	Qualifies	59%
Chip	NVIDIA, AMD, AVGO	High (NVIDIA)	**Specifies**	66%

Source : Company filings, TrendForce, JP Morgan

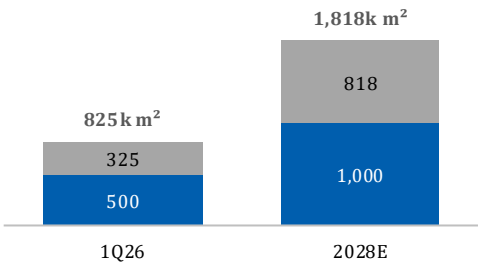
Figure 2B: Rival T-glass Qualification Ladder

Date	What Nittobo management said about rival T-glass
Aug-24	Several rivals sampling with customers; none adopted yet
May-25	Rivals now making product to customer specifications
Aug-25	Taiwanese rivals start qualification; quality and volume not yet adequate
Nov-25	Taiwanese rivals reach spec-in (designed into customer products); volume still short
May-26	"Some movement of new entrants observed"
Aug-26	Package makers starting to adopt non-Nittobo low-CTE glass, as Nittobo capacity is not yet in place

~2 years from sampling to spec-in, during the shortage

Source : Company filings, Team Analysis

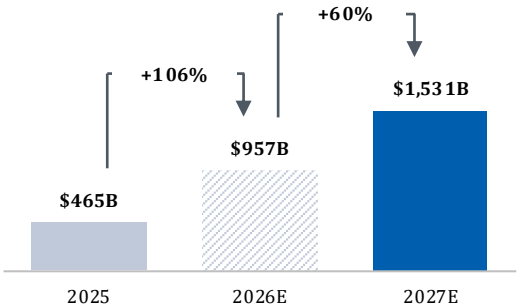
Figure 2C: T-glass Capacity by Producer (k m²)



2026–28E: Nittobo +100% · Rivals +152% · **Total +120%** · CoWoS +100%

Source : Prismark, Team Analysis

Figure 2D: Top-9 CSP Capex ($B) and YoY Growth



Source : TrendForce, Google, AWS, Meta, Microsoft, Oracle, ByteDance, Tencent, Alibaba, Baidu

Figure 2E: Substitution Milestones

	Low-Dk → Quartz	T-glass → Glass Core
2026	M9 Q-glass in Rubin midplane and LPX boards	Absolics and Samsung EM samples in customer qualification
2027	Rubin Ultra (Kyber) 78+ layer M9+ boards	Samsung EM glass core (2H); Absolics commercial start
2028	M10 CCL in qualification; Nittobo launches next-gen Vlex	Samsung EM glass substrates; TSMC CoPoS mass production (2H)
2030	Q-glass stays niche at ~40x E-glass cost	Intel and TSMC glass substrates at scale

Source : TrendForce, The Elec, DIGITIMES, Company filings

Value Chain & Profit Pools

Margin Stacks Downstream: Nittobo's Electronic Materials segment earned a 39% operating margin in FY26A, well below Ajinomoto's ABF film (54%), TSMC's CoWoS (59%) and NVIDIA (66%) (**Figure 2A**). All four layers are close to single-source, yet Nittobo's margin sits nearer to the laminate makers it sells to (EMC 27%) than to the other near-monopolies.

Pricing Power Sits Further Down the Chain: Nittobo negotiates prices once a year with laminate makers, whose own margins are thin, rather than with the chipmakers that value its cloth most. Downstream, pricing is already doing the work: ¥26.5B of Ibiden's ¥35B electronics profit upgrade came from price and mix against ¥3B from volume, and Ajinomoto's ABF margin rose from 48% to 57% in FY26A. Nittobo has chosen cost-recovery pricing instead.

Competitive Landscape & Qualification

Buyer ≠ Chooser: Three groups shape Nittobo's business (**Figure 2A**). NVIDIA, AMD and the hyperscalers specify the grade and lobby for allocation, and hyperscalers booked most of Nittobo's premium cloth for 2026 (Nikkei, Aug-26). Substrate makers and TSMC qualify each supplier, but laminate makers place the orders and pay. Because the specifiers care more about secure supply than about price, a shortage pushes them to approve a second source.

A Moat That Has Already Been Crossed: A new cloth must pass laminate, substrate, packaging and chip-designer testing, which takes one to two years, and this is the main defence in the bull case. Management's own comments show the clock has already run: rivals were sampling in Aug-24, reached spec-in by Nov-25, and package makers began adopting non-Nittobo low-CTE glass in Aug-26 because Nittobo could not supply enough (**Figure 2B**). Once a rival is designed in, it keeps the slot after the shortage ends.

T-glass, Dominant Today but Diluting by 2028: Nittobo holds ~90% of T-glass, with 500k m² of capacity against rivals' 325k m² in 1Q26. Taiwan Glass has been in mass production since 1Q26, while Grace, Taishan and Nan Ya are not yet qualified. Prismark expects rival capacity to grow ~2.5x by 2028, cutting Nittobo's share of capacity to ~55% (**Figure 2C**).

Low-Dk, Already Contested: Nittobo supplies 60–70% of NE-glass, alongside Asahi Kasei, Taiwan Glass, Fulltech and Taishan. Taiwan Glass is spending NT$2.24B to lift its high-grade lines from 4 to 12, and customers already use NE-glass in place of T-glass.

Competition Signposts to Watch

Signpost	Where it stands (Oct-26)	What would confirm our view
A customer names a second T-glass supplier	Spec-in Nov-25; adoption starting Aug-26	A named customer
Taiwan Glass monthly glass-fibre sales	In mass production since 1Q26	Sustained month-on-month growth
Grace / Taishan / Nan Ya T-glass qualification	Not yet qualified	Any one qualified
Asahi Kasei quartz cloth qualification	Due by Mar-27	Qualified on schedule

Supply vs Demand

A Cobweb Industry: Glass-fibre capacity is ordered on today's price and arrives about two years later, as furnaces take over two years to build and looms carry 18–24 month lead times. Producers therefore add at the same time: Nittobo is tripling T-glass, Taiwan Glass is lifting high-grade lines from 4 to 12 and Nan Ya is adding weaving, all ramping in 2H26–27.

Capex Growth Is Slowing: Top-9 CSP capex rises 106% to $957B in 2026E and 60% to over $1.5T in 2027E, but UBS sees only 12% growth in 2028E (**Figure 2D**). Cloth orders follow the growth in capex, not its level, so orders can fall before spending does.

Funded Increasingly by Debt: Hyperscaler capex exceeds operating cash flow from ~3Q26 (Epoch AI), only Microsoft among the six largest spenders is FCF-positive in 2027E (S&P Global), and ~$1.5T of 2025–28 data-centre spend is expected to be externally financed, mostly through private credit (MUFG).

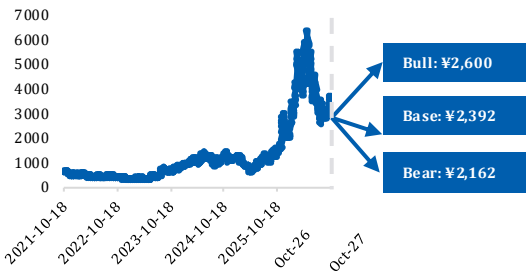
Substitution Timeline

Quartz Is the Near-Term Substitute: 224G and 1.6T links push high-end boards to M8.5–M9 laminates, which mainly use quartz cloth, already found in parts of NVIDIA's Rubin. At ~40x the cost of E-glass, quartz should stay on top-grade boards (Shin-Etsu, Asahi Kasei, Glotech), but there it competes directly with Nittobo's Low-Dk. Asahi Kasei's qualification, due by Mar-27, is the signpost inside our horizon.

Glass Core Is a Multiple Risk, Not an Earnings Risk: Glass-core substrates would replace the T-glass core, but timelines keep slipping. Absolics moved from 2H24 to 2027, Samsung EM targets glass core in 2H27 and substrates in 2028, TSMC plans CoPoS mass production in 2H28 with glass after 2030, and Intel is ~2030 **(Figure 2E)**. This falls beyond our 12-month horizon and affects the multiple the market pays, not FY27–28E earnings.

Appendix 3: Valuation & Financials

Figure 3A: Stock Price under Bull/Bear Scenarios (¥)



Source : Yahoo Finance, Team Analysis

Bull Case: Our bull case assumes the T-glass shortage persists through the forecast period, with Nittobo extending its capacity expansion beyond the current plan and demand from AI package substrates absorbing every new phase, keeping furnaces fully utilised. As each phase comes online, Nittobo passes on part of the investment burden through capacity-linked price increases, while Low-Dk prices firm as NER-glass gains adoption in higher-grade CCLs. The additional capex is matched by higher volume, so the rising depreciation is absorbed and EM margins expand above current levels.

Bear Case: Our bear case assumes the T-glass market rebalances as Nittobo's committed capacity comes online, with customers dual-sourcing from Taiwanese and Chinese competitors and utilisation falling steadily from FY28E. As competing supply expands, prices for both T-glass and Low-Dk decline modestly from FY29E. With capex already committed under the Medium-Term Plan, depreciation rises in full against a slower-growing revenue base, compressing EM margins well below Street estimates.

Figure 3B: Cost of Debt

Credit Rating Approach	
Rf Rate – JGB 10Y	3.0%
Actual Credit Rating (R&I)	A-
Default Risk Spread	1.25%
Pre-Tax Cost of Debt	4.25%
Tax Rate	29.0%
After-Tax Cost of Debt	3.02%

Source : Yahoo Finance, Company filings, Team Analysis

Figure 3C: Cost of Equity

Capital Asset Pricing Model	
Rf Rate – JGB 10Y	3.0%
Relevered Beta	1.15x
Equity Risk Premium	5.14%
Cost of Equity	8.9%

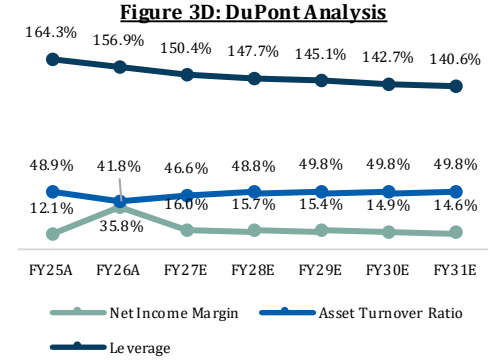
Source : Yahoo Finance, Team Analysis

Valuation	Weightage	TP (¥)
P/E	25.0%	2,317
P/B	25.0%	2,771
EV/EBITDA	25.0%	2,868
EV/Rev	25.0%	1,610
DCF – Perpetuity	0.0%	1,447
DCF - Terminal Multiple	0.0%	1,835
Implied share price	100.0%	**2,392**

Discounted Cash Flow Analysis

DCF was conducted in order to understand the intrinsic value of Nittobo, but it was not used in the weighted valuation due to the characteristics of the semiconductor industry, which is cyclical and therefore uses multiples to justify the values. Our team used the Japanese 10 Year government bond rate and Nittobo's credit rating of A-, which translates to default risk spread of 1.25%, according to Damodaran, which leads to pre-tax cost of debt of 4.25% and post-tax cost of debt of 3.02% (**Figure 3B**). Nittobo's cost of equity was calculated using the Capital Asset Pricing Model (CAPM), resulting in an estimated Cost of Equity of 8.9%. This was derived using the 10-Year Japanese Government Bond (JGB) rate of 3.0% as the risk-free rate, a levered beta of 1.15x, and an equity risk premium of 5.14% (**Figure 3C**)

Figure 3D: DuPont Analysis

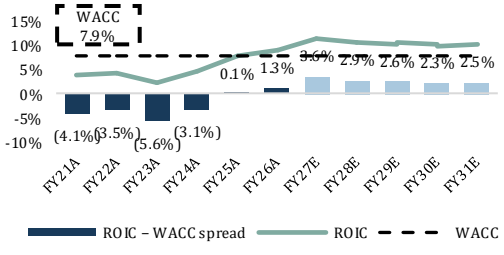


Source : Company filings, Team Analysis

Financial Analysis

DuPont Analysis: Nittobo's net profit margin is stable throughout, except for FY26A when Nittobo sold its Yaesu (Tokyo) land to Sumitomo Realty for about ¥39B in cash, against a book value of only about ¥5B, which produced a ¥34.2B pre-tax gain and lifted the FY26A net profit margin by about 20pp, from roughly 15% to the reported 35.3% (**Figure 3D**). Moving forward, our team projects the net profit margin to slightly decrease from 16% to 14.6% in FY27-31E, as depreciation from the capex cycle builds, and the equity multiple falls from 1.50x to 1.41x as retained earnings grow the equity base faster than debt. Asset turnover rises only modestly to about 50%, and then flattens, so new capacity adds assets roughly in line with revenue rather than lifting efficiency. Overall, Nittobo's ROE is around 10-11%, which is ahead of the MTP target of ≥8%, but this is nowhere close to the ~20% sustainable ROE that a ~3x P/B multiple implies.

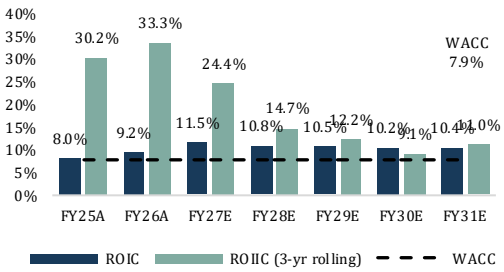
Figure 3E: ROIC – WACC Spread



Source : Company filings, Team Analysis

Profitability Metrics: Nittobo's FY21-24A earnings were below its cost of capital, as ROIC was 2-5% against a 7.9% WACC, resulting in negative spread. This was the last capex cycle when Taiwan capacity and new lines came on just as energy costs spiked. It only crossed WACC in FY25-26A, as T-glass and special-glass pricing surged through asset base that was already depreciated. Those two years' ROIIC of 30-33% reflects a large NOPAT increase earned on very little new capital. We forecast ROIC to peak at 11.5% in FY27E, then fades to about 10%, so the ROIC-WACC spread narrows from 3.6% to around 2.3-2.5%, meaning that extra investment earns less (**Figure 3E**).

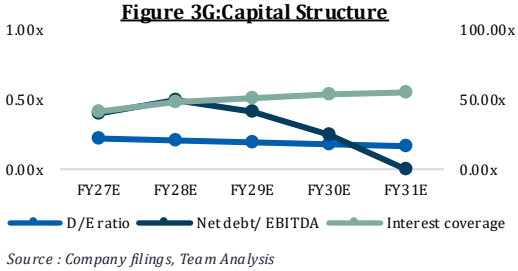
Figure 3F: ROIC vs ROIIC



Source : Company filings, Team Analysis

3-year rolling ROIIC (return on incremental invested capital) falls from 24.4% in FY27E to 9.1% in FY30E as it consistently approaches WACC of 7.9% (**Figure 3F**). The high ROIIC in FY25-26A is before the ¥120B capex programme, while demand drove up the pricing and utilisation rate of the existing facilities. The expansion is therefore adding capital that, in the future, barely earns the cost, as new lines carry full depreciation and start-up costs against prices that only pass costs through. The 11.0% uptick in FY31E is a denominator effect: capex slows down, so invested capital stops growing. Overall, the growth coming from the aggressive expansion plan is minimal as the additional capital doesn't translate to incremental returns. That's at odds with a stock at ~3x book and a premium multiple, which only make sense if FY25-26A style incremental returns continue. Our numbers show those returns converging toward WACC, as done in the last cycle.

Figure 3G:Capital Structure



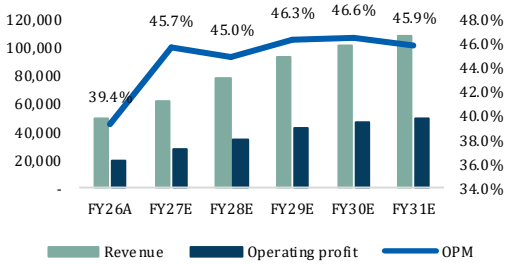
Source : Company filings, Team Analysis

Capital Structure: Debt/equity ratio falls from 23.1% in FY27E to 16.3% in FY31E. This leverage reduction is primarily driven by organic growth in Nittobo's equity denominator, fuelled by sustained retained earnings from electronic materials. As the capital structure shifts towards a higher proportion of equity, the WACC will gradually converge towards our cost of equity figure of 8.9%. Net debt/EBITDA falls from 0.40x to 0.01x, which is conservative, requiring fewer years of current operating earnings to pay off debt. Moreover, interest coverage ratio rises from 40.83x in FY27E to 55.46x in FY31E, implying an increased ability to service debt. Overall, we see this as a sign of strong financial health for Nittobo.

Appendix 4: Opposite Pitch

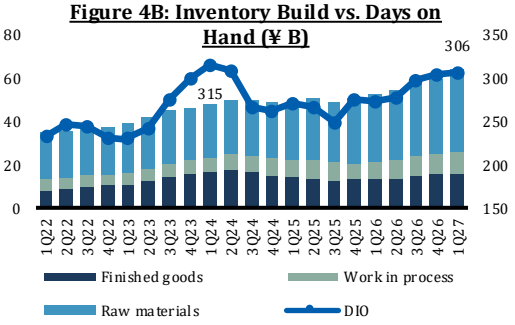
Why has the market not factored in our investment thesis and reached our target price yet?

Figure 4A: Numbers at Street's Case (¥)



Source: Company filings, Team Analysis

Figure 4B: Inventory Build vs. Days on Hand (¥ B)



Source: Company filings, Team Analysis

The Margin Narrative Will Only Realise in 4Q27: Consensus is still optimistic on Nittobo attaining mid-40s OPM through FY29E as they believe that shortage will persist past the capacity expansion (**Figure 4A**). However, we foresee a market correction due in 4Q27 when management provides guidance on new depreciation tied to capex investments and lower margins. This is evidenced in our analysis that 1) the market is not pricing in the new ramp and start-up cost associated with expansion and rising depreciation, and 2) customers are ordering ahead of use, as seen in their rising inventory on hand.

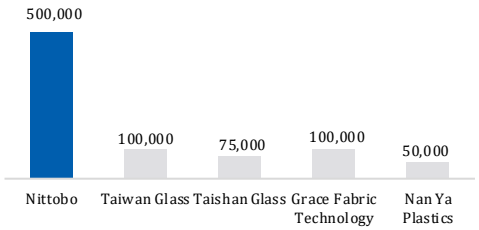
The Market is Waiting for Evidence: The re-rating should come in stages as proof arrives. First are the 2Q results and the 2H EM margin guide (Nov 2026), then the Fukushima start-up (early 2027), Asahi Kasei's quartz qualification (~Mar 2027) and the FY28E guidance (May 2027). Each event tests pricing power and narrows the gap between the share price and our view, well before the full earnings hit in FY28–29E.

The Coverage Blindspot: Nittobo is primarily covered by equity analysts who look mostly at Japanese domestic disclosures, where its +¥13.2B inventory accumulation is viewed favourably as pre-stocking for upcoming capacity (**Figure 4B**). However, upstream and downstream evidence in Taiwanese filings shows that Copper Clad Laminate (CCL) makers are already experiencing an inventory glut, which suggests that the supply chain is overstocked (**Figure 17**). We posit that this information asymmetry contributes to the delayed stock reaction.

Slower Growth is Priced In, but Falling Profits are Not: The shares have already fallen ~53% from their peak, so the market expects growth to slow. What it has not priced is orders falling sharply just as a much larger fixed-cost base comes online. Our FY23A calibration shows ~¥50–55 of operating profit lost for every ¥100 of lost sales, and that earnings hit is where our downside comes from.

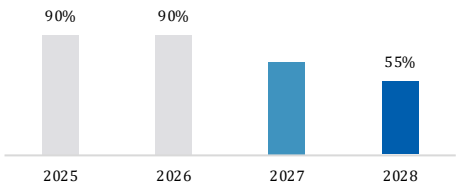
If someone makes an opposite pitch, what would be the thesis for them and why doesn't the opposite thesis negate our thesis?

Figure 4C: Estimated Market Capacities (m^2/Q1 2026)



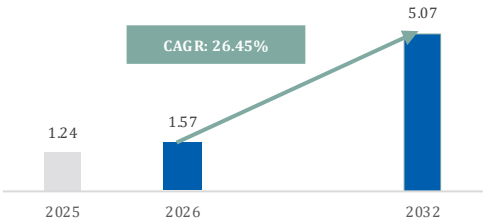
Source : Prismark, Team Analysis

Figure 4D: Nittobo's Market Share (Est.)



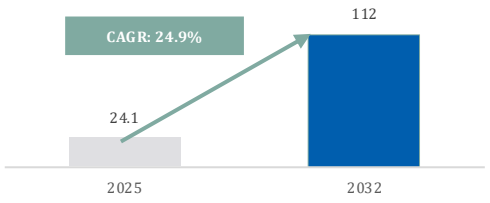
Source : Prismark, Team Analysis

Figure 4E: Glass Core Substrate Market ($USD billions)



Source : Mordor Intelligence

Figure 4F: Quartz Cloth Market ($USD millions)



Source : Valuates Reports

Counter-Thesis: Nittobo is The Dominant Player and Deserves a Premium

Competitors Are Still Lagging Behind: According to estimates from Prismark Partners, Nittobo's T-glass production capacity is the largest in the industry at 500,000 square metres in 1Q26 (**Figure 4C**) and is expected to double by 4Q28. Industry sources note that while competitors are also undergoing T-glass technology evaluations with big tech, the technology gap versus Nittobo remains substantial. The core challenge in electronic glass fibre is the extreme precision required to draw molten glass into continuous filaments down to 3.5 microns in diameter without breakage. Manufacturers in Taiwan and China still fall short of big tech's technical qualification standards, leaving Nittobo in a position of effective market monopoly.

GCS is a Long-Term Horizon Concept, Not a Near-Term Threat: Glass Core Substrates (GCS) still faces unresolved yield and thermal stress bottlenecks at scale. For the foreseeable cycle, woven T-Glass remains the only viable commercial standard for high-density interconnect (HDI) and GPU substrate packing.

Quartz Cloth Will Not Threaten Nittobo's T-Glass: Drawing pure quartz into uniform, fine fibres for weaving without high breakage rates remains a challenge. The cost structure of quartz cloth is also radically higher, creating massive adoption friction unless performance demands absolutely mandate it. Lastly, integrating quartz-based layers into HDI PCBs and CCL introduces severe thermal expansion, processing and lamination hurdles that the supply chain is still solving.

Pricing Power Should Persist Amidst Shortage: Nittobo's pricing power in T-Glass should be immense enough during current short supply conditions that it can pass higher costs directly downstream to substrate giants without hurting gross margins.

Our Rebuttal: A Premium Doesn't Break Our Thesis

We Agree On The Premium. We Disagree On Its Extent: Nittobo's dominance in electronic materials justifies trading above the sector. What we dispute is the stock's relative premium across all its operating segments. **(Figure 29)**. Nittobo currently trades at a premium against industry peers on both P/E and EV/EBITDA, even across segments that it does not monopolise. We believe this to be due to the spillover effects of the AI narrative.

The Monopoly Effect is Peaking: As competitors continue to expand capacity, Nittobo's market share is projected to fall to approximately 55% **(Figure 4D)**. We believe the Street overly fixates on the supply-demand dynamics of glass fibre while overlooking Nittobo's market share dilution and growing competitive pressure.

Alternatives Are Growing Quickly: GCS is already a relatively large market valued at US$1.57B, and projected to grow at a 26.45% CAGR through 2032 **(Figure 4E)**. Additionally, the quartz cloth market is due to grow at a 24.9% CAGR through 2032 **(Figure 4F)**. T-glass faces a multi-front threat to its long-term addressable market share and pricing power as alternatives ramp up.

Pricing Power is Constrained by Downstream Substrate Concentration: While Nittobo enjoys pricing leverage during supply crunches, its primary direct buyers consist of a tightly consolidated group of substrate giants (e.g., Unimicron, Ibiden, Nan Ya, EMC). These major customers possess immense long-term volume leverage and are actively pushing hyperscalers to qualify secondary sources, establishing a structural cap on long-term gross margin expansion.

Appendix 5: ESG

Figure 5A. Management's FY30 Targets

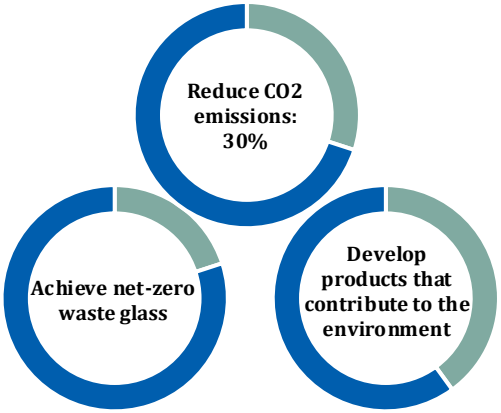
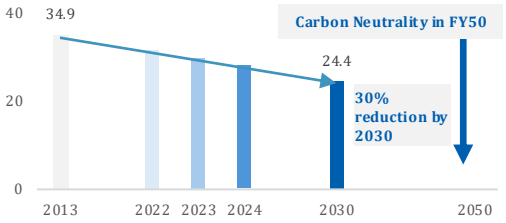
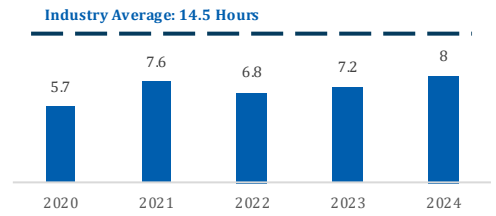


Figure 5B. CO2 Emissions



Source : Company filings, Team Analysis

Figure 5C: Average Monthly Overtime Hours



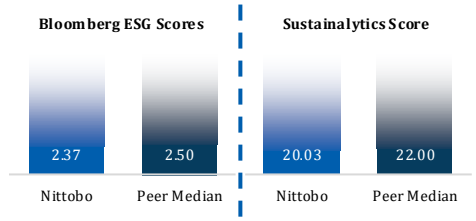
Source : Company filings

Figure 5D. Board of Directors



Source : Company filings

Figure 5E. ESG Scores Against Peers



Source : Bloomberg

Environment, Social, Governance

Nittobo's ESG framework is already mature **(Figure 5A)**, we expect that the market is unlikely to pay a further premium for it. The ESG risks we identify primarily sit in 1) the emissions and Scope 3 impact of capacity expansion, and 2) the labour constraint at the Fukushima production base. We see ESG as a supporting case to our **SELL** recommendation, with the capex cycle and labour constraints being the main monitorable.

ESG Performance Analysis – Environment (E)

Absolute Emissions to Rise With Capex: Scope 1 and 2 emissions fell to 269k tonnes in FY24A, 22.9% below FY13A. Carbon intensity fell 37.5% to 2.47 t-CO2/¥M since FY20A, but this ratio is flattened by sales growth attributed to strong Special Glass sales and the weak yen. We believe some of the improvement reflects higher-value revenue mix and currency translation rather than actual operational decarbonisation. Absolute emissions fell only 13.5% over the same period, and the FY30 target of a 30% reduction still requires a further 9% cut, exacerbated by increased emissions from capacity expansion as warned by management.

Decarbonisation is Largely Unproven: Electric and hybrid furnaces, solar at Fukuyama, CCUS (Carbon Capture, Utilisation, and Storage), alternative fuels, and an e-methane pilot with Koriyama City and partners commencing FY26A, are all early-stage endeavours. Management notes that hydrogen and next-generation fuels cannot be solved by Nittobo alone, citing procurement costs and operational dependencies. We believe the FY50 carbon-neutrality target to be speculative, and easier said than done **(Figure 5B)**

ESG Performance Analysis – Social (S)

Talent is Scarce: The CEO identifies recruitment in Fukushima Prefecture (high youth outflow) and a thin middle-management layer as key risks to capacity expansion. Nittobo currently targets a 20% increase in global headcount by FY27. We see a mismatch between a demand-led capacity ramp and a difficult regional hiring market. In attracting talent, we foresee that Nittobo will have to continually introduce additional incentives. Bonuses were introduced in FY24A and triggered when operating profit targets are met in the same year. Qualification subsidies of up to ¥500k and genba-leader training were initiated to support skills transfer in manufacturing.

Workstyle and Health: Average overtime of 8.0 hours a month is 45% below the manufacturing average of 14.5 hours, with no breaches of statutory limits. However, it has risen from 5.7 to 8 hours **(Figure 5C)** and we expect it to rise further as production ramps.

ESG Performance Analysis – Governance (G)

Governance Structure is Strong: We see governance as the strongest pillar of Nittobo's ESG efforts. The board has seven members **(Figure 5D)**, four of them external (57%). All three statutory committees are chaired by external directors. Director attendance was 100% at 12 of the 13 Audit Committee meetings, and board effectiveness came in at an average 4+/5 as evaluated by external attorneys. Pay is also performance-linked to operating profit and ROE, with performance-linked stock to spur performance.

Why ESG Reinforces Our Thesis

The Bloomberg ESG score evaluates how well a company managed financially material industry-specific risks and opportunities, using a scale of 0 to 10. Weightages by element vary by industry materiality.

While Nittobo's ESG disclosure is well-established, its Bloomberg ESG score of 2.37 remains lower than industry peers at 2.50 **(Figure 5E).** This is heavily driven by gender wage gaps and low female leadership representation, as evidenced by its lagging score on Labour & Employment Practices. Overall, Nittobo ranks below median for three out of five material ESG elements. Additionally, Nittobo has a Sustainalytics Score of 20.03/100, indicating moderate unmanaged ESG risk. Addressing these concerns will require incremental compliance investments from Nittobo, creating further margin friction as competition in T-glass intensifies, and contributing to our overall **SELL** recommendation.

Sustainable Product Score:	Social Supply Chain Management:	Labour & Employment Practices:	Board Composition:	Data Security & Customer Privacy:
Above Median: 2.0/10.0 **Weightage:** 27%	**Median:** 1.0/10.0 **Weightage:** 16%	**Lagging:** 1.2/10.0 **Weightage:** 11%	**Below Median:** 5.0/10.0 **Weightage:** 10%	**Below Median:** 2.0/10.0 **Weightage:** 9%

Source : Bloomberg